

HVDC INTERCONNECTS

Design Basis, Design Basis Accident, and Military Action Extension

DBA-ES-TX-FC3: HVDC Interconnect Facility Class

A Child Document Under
DBA-ES: Electric Sector Design Basis Framework (v0.6)

Version 0.1 — DRAFT

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For review by TransCo engineers, NERC reliability analysts, FERC staff, and policy officials responsible for electric sector critical infrastructure protection.

Revision History

Version	Date	Author	Description
0.1	June 2026	T. Roxey	Initial working draft. First publication of the DBA-ES-TX-FC3 facility-class instrument under DBA-ES sector parent (v0.6). Established facility class definition (point-to-point HVDC, $\geq 2,000$ MW, cross-interconnection role). Introduced TCI (Transfer Capability Isolation) as the governing FC3 consequence-duration metric. Retained RT_HVLLC for converter transformer replacement as a secondary metric. Instantiated all four sector-level postulated event categories with FC3-specific events. Developed DBA non-adversarial consequence envelope. Applied DB→DBA→DBA-MA three-layer progression. Incorporated the Nelson River system as the consequence model and the 2003 Northeast Blackout as the cascade precedent.

Section 1 — Purpose, Scope, and Inheritance

1.1 Purpose

This document establishes the Design Basis for high-voltage direct current (HVDC) interconnects within the Electric Sector Design Basis Series. It is the third facility-class instrument in the series, following DBA-ES-HY-FC1 (large hydroelectric storage facilities) and DBA-ES-TX-FC2 (large EHV transmission substations).

The document applies the DB→DBA→DBA-MA three-layer analytical progression to HVDC interconnects serving a cross-interconnection transfer role: a baseline Design Basis (DB) establishing postulated events from equipment failure, natural hazards, and operational conditions; a Design Basis Accident (DBA) layer bounding the non-adversarial consequence envelope; and a military action extension (DBA-MA) applying the four characteristics of deliberate adversarial action — intentionality, persistence, precision, and denial — to each postulated event category.

HVDC interconnects occupy a unique position in the electric sector's consequence structure. Unlike all other facility classes in the DBA-ES series, whose consequences propagate within a single interconnection, HVDC interconnects carry consequences across interconnection boundaries. The loss of a major HVDC link removes the transfer capability that allows one interconnection to support another under stress — a consequence with no analog in AC transmission and no precedent in the non-adversarial design basis frameworks that govern electric sector planning. This cross-interconnection isolation function defines the FC3 class and governs its consequence analysis.

This document introduces Transfer Capability Isolation (TCI) as the governing consequence-duration metric for FC3. TCI is defined as the duration during which cross-interconnection transfer capability is unavailable following loss or block of the HVDC link, expressed in hours, days, or months, depending on the initiating event. TCI complements the RT_HVLLC metric established in DBA-ES §6.2: RT_HVLLC governs the equipment replacement timeline following physical damage to converter transformers; TCI governs the transfer capability consequence regardless of whether physical equipment damage is involved.

1.2 Facility Class Definition

A facility within the scope of DBA-ES-TX-FC3 must satisfy all three of the following criteria.

Technology class. The facility is a high-voltage direct current transmission system using line-commutated converter (LCC) or voltage-source converter (VSC) technology. Both LCC and VSC HVDC systems are within scope; specific postulated events and acceptance criteria may differ between the two technologies and are noted where relevant.

Transfer capacity. The HVDC link's nameplate transfer capacity equals or exceeds 2,000 MW in either transfer direction. This threshold is calibrated to the scale at which HVDC link loss produces measurable frequency and voltage transients in the receiving AC system that challenge the system's frequency response capability. HVDC links below this threshold may still be consequential in small or weak AC systems, but are not within the scope of this instrument.

Cross-interconnection role. The HVDC link connects two separate AC interconnections or sub-regions whose AC systems are not synchronously coupled. In North America, this includes links between the Eastern Interconnection, Western Interconnection, and ERCOT, as well as links to Canadian provincial systems operating asynchronously. HVDC links connecting two nodes within the same synchronous interconnection — back-to-back HVDC — are within scope if they satisfy the capacity threshold and serve a regional isolation-prevention function equivalent to a cross-interconnection link.

1.3 The TCI Metric

Transfer Capability Isolation (TCI) is introduced in this instrument as the primary consequence-duration metric for FC3. It is defined as:

TCI: The continuous duration, measured from the initiating event, during which the HVDC link is unable to deliver its design transfer capacity in the governing transfer direction, expressed in hours (for operational events), days (for equipment repair events), or months (for replacement-class events).

TCI is a functional metric, not a physical one. It measures the loss of the transfer capability function, not the loss of specific equipment. TCI begins when the HVDC link is blocked, tripped, or otherwise unable to deliver design transfer capacity, and ends when full transfer capability is restored. TCI is independent of the cause of the interruption: a control system block, a converter

transformer failure, a DC line fault, or a military precision strike, all of which engage TCI from the moment of blockage.

TCI is the FC3 analog of RT_HVLLC in DBA-ES-TX-FC2. Where RT_HVLLC measures the duration of a facility's physical unavailability following destruction of its most difficult-to-replace component, TCI measures the duration of the transfer capability function's unavailability regardless of physical cause. For FC3, the most severe TCI scenario is one in which TCI equals RT_HVLLC — that is, a physical attack on the converter transformer that makes the link unavailable for the full component replacement period of twelve to eighteen months.

1.4 Evidentiary Anchors

Two anchors establish the evidentiary foundation for FC3.

Nelson River HVDC System (Manitoba, Canada). The Nelson River bipolar HVDC system — comprising Bipole I (1,620 MW, commissioned 1970–1977), Bipole II (2,000 MW, commissioned 1978–1985), and Bipole III (2,000 MW, commissioned 2021) — is the primary reference case for cross-interconnection HVDC consequence analysis in North America. The system carries hydroelectric generation from northern Manitoba to the Winnipeg load center and interconnects with the Manitoba Hydro system's exports to Minnesota and other US markets. The prolonged debate over Bipole III routing — in which the decision to route the line west of Lake Winnipeg rather than east was driven by concerns about the consequence of a single transmission corridor fault taking down both Bipole I and Bipole II simultaneously — is a documented case study in HVDC corridor concentration risk and the policy response to it. The Nelson River system establishes the consequence model: a major HVDC system serving a regional load center and a cross-border export function, whose loss requires emergency imports or generation backstop from remote sources.

2003 Northeast Blackout (August 14–15, 2003). The 2003 Northeast Blackout is the definitive evidentiary anchor for cross-interconnection cascade dynamics. The initiating events were modest — line contacts in Ohio, compounded by software failures in FirstEnergy's EMS — but the cascade propagated across multiple control areas. It ultimately separated a large portion of the Eastern Interconnection. The blackout demonstrated that the loss of transfer capability between regions, under conditions of prior constraint, transforms a local contingency into a regional cascade. The final NERC/DOE investigation report established the sequence by which information loss (CE-type events) compounded physical events (PE-type events) to produce consequences beyond the sum of their individual effects. This interaction pattern — the compound consequence of simultaneous physical and communications degradation — is the primary analytical framework for FC3's CE consequence analysis.

1.5 Inherited Methodology

All sector-level methodology is inherited from DBA-ES: Electric Sector Design Basis Framework (v0.6). The following table identifies inherited elements and FC3-specific adaptations.

Element	Source	FC3-Specific Adaptation
DB→DBA→DBA-MA three-layer progression	DBA-ES §3.1	Inherited without modification
Three required adaptations	DBA-ES §3.2	Inherited; acceptance criteria include transfer capability role
Postulated event categories: PE, SE, CE, XC	DBA-ES §3.5	FC3-specific events instantiated in Section 3
Bounding event methodology	DBA-ES §4.1	Converter transformer is the governing long-lead-time equipment; TCI is the primary consequence-duration metric.
Natural hazard compound-event requirement	DBA-ES §4.3	Geomagnetic disturbance + seismic as governing compound case
Three-tier consequence taxonomy	DBA-ES §6.1	Cascade pathway governs; TCI defines Tier 2 duration; Tier 3 through regional isolation.
Long-lead-time equipment and an extended consequence regime	DBA-ES §6.2	RT_HVLLC secondary to TCI; converter transformer lead times 12–18 months
MA extension trigger criteria	DBA-ES §1.5	FC3 satisfies Triggers 1, 2, and 3 by definition; Trigger 5 (corridor significance) also applies
Four MA characteristics	DBA-ES §5.2	Applied to each FC3 postulated event category in Section 5
Dual Design Basis architecture	DBA-ES §3.4	Physical DB (this document) + AI Governance DB interaction acknowledged in Section 3.6

1.6 Position in Hierarchy

Document Number: DBA-ES-TX-FC3-001

Series: Electric Sector Design Basis Series — Facility Class Instruments

Parent Document: DBA-ES: Electric Sector Design Basis Framework (v0.6)

Position in hierarchy: Facility-class instrument. All sector-level methodology is inherited from DBA-ES. This document applies that methodology to HVDC interconnects and introduces the TCI consequence-duration metric specific to this facility class.

Section 2 — Facility Profile: HVDC Interconnect

2.1 Physical Configuration

A point-to-point HVDC interconnect consists of two converter stations — a rectifier (AC-to-DC conversion) and an inverter (DC-to-AC conversion) — connected by a DC transmission line or cable. Each converter station contains the converter transformers, thyristor or IGBT valve assemblies, smoothing reactors, DC filters, AC filters, and the control and protection systems that manage the AC-DC conversion process. The converter stations are substantial fixed installations; a major HVDC converter station for a 3,000 MW system may occupy ten to twenty acres and require hundreds of miles of DC transmission infrastructure.

The converter transformer is the critical element that defines the FC3 consequence profile, analogous to the large power transformer in FC2. Converter transformers are custom-designed for HVDC service: they must withstand DC voltage stress on the valve-side windings, harmonic currents, and the electrical stresses unique to the commutation process. These requirements produce a design that is distinct from standard power transformers and that cannot be substituted with a standard unit. The custom design requirement, combined with the limited global manufacturing capacity for large HVDC converter transformers, produces the RT_HVLLC characteristic that makes physical damage to the converter transformer the governing consequence event.

The DC transmission line or cable is the physical infrastructure connecting the two converter stations. Overhead DC lines are similar in construction to AC transmission lines but operate at higher voltage-to-ground ratios and carry DC rather than AC. They share the physical vulnerability characteristics of overhead AC lines — susceptibility to conductor damage, insulator flashover, and tower failure — but the consequences of DC line faults differ from AC line faults because DC line protection operates differently and the restoration sequence is more complex.

2.2 Grid Topology Role

An HVDC interconnect in a cross-interconnection role performs a function that has no AC equivalent: it allows power to flow between two AC systems that are not synchronously coupled and therefore cannot be directly interconnected by AC transmission. This asynchronous transfer function provides three grid services simultaneously: it enables energy exchange between interconnections with complementary supply and demand profiles; it provides a transfer capability resource that either interconnection can call upon under stress; and it provides a degree of electrical isolation between the two interconnections, preventing fault propagation across the DC link in either direction.

The isolation function is both an asset and a vulnerability. As an asset, it means that a fault in one interconnection cannot directly propagate to the other through the HVDC link — the

converters block before the fault current can transfer. As a vulnerability, it means that the loss of the HVDC link removes the transfer capability that one interconnection was depending on, without the losing interconnection having any electrical path to recover it. When the Nelson River Bipole I was forced out of service, Manitoba could not receive power from the Eastern Interconnection through an alternate AC path — there is no such path. The HVDC link is the only transfer mechanism, and its loss is absolute in a way that AC line loss is not.

2.3 Regulatory Context

HVDC interconnects are subject to NERC reliability standards as Transmission Owners and Transmission Operators, including the FAC (Facilities Design, Connections, and Maintenance) standards and the applicable TPL reliability standards. However, HVDC systems present regulatory classification challenges because their operating characteristics — instantaneous power control, lack of natural reactive power contribution, absence of synchronous inertia in VSC systems — do not map cleanly onto the AC transmission standards that NERC developed for synchronous generators and AC lines.

No mandatory physical security standard equivalent to NERC CIP-014 has been established specifically for HVDC converter stations, although major converter stations may be designated under CIP-014 based on their consequence analysis. The absence of a mandatory physical security standard for HVDC infrastructure as a distinct asset class is an identified regulatory gap. This document supplies the design basis analysis that a physical security standard for HVDC would need to draw upon.

FERC has jurisdiction over HVDC facilities that cross international borders or interconnect separate control areas under the Federal Power Act. Specific FERC orders applicable to HVDC systems include the interconnection standards and the applicable market rules governing DC transmission capacity and scheduling.

Section 3 — Design Basis: FC3 Postulated Event Categories

3.1 Sector-Level Inheritance

The four sector-level postulated event categories — PE (Power Supply and Grid Interface), SE (Structural and Physical Integrity), CE (Control and Communications), and XC (Cross-Sector Cascade) — are inherited from DBA-ES §3.5. Each is instantiated below with FC3-specific events. No additional postulated event categories are locally defined for FC3; the four inherited categories are sufficient to bound the FC3 event space.

3.2 Category PE — Power Supply and Grid Interface Events

Power supply and grid interface events at FC3 are primarily events that affect the AC system conditions at one or both converter terminals, the DC line connecting the terminals, or the

auxiliary power supply at the converter stations. Unlike FC2 (where the facility is a passive power delivery node), an HVDC converter station is an active power conversion device whose operating state depends continuously on both AC system conditions and DC line conditions. Events affecting either dimension can force a converter block.

Code	Event	Initiating Condition	Governing Concern
PE-1	AC system fault causing forced converter block.	A fault on the AC system at either the rectifier or inverter terminal causes the HVDC control system to block the converters as a protective response. The HVDC link ceases power transfer instantaneously. AC protection systems clear the fault; the HVDC link may restart automatically following fault clearance.	The instantaneous loss of the HVDC transfer block — potentially several thousand megawatts — initiates a frequency and voltage transient in both the sending and receiving AC systems simultaneously. The AC systems must absorb the transfer loss within their frequency response capability. TCI begins at the block and ends at the restart.
PE-2	Commutation failure at the inverter terminal	Commutation failure occurs when the inverter thyristors fail to transfer current correctly, typically caused by a voltage depression on the AC system at the inverter terminal. Commutation failure momentarily collapses DC and may propagate to sequential commutation failures if the AC voltage depression persists.	Repeated commutation failures can force a sustained block of the HVDC link even without physical equipment damage. The governing concern is whether commutation failure cascades into a full link shutdown and how quickly the AC system voltage recovers sufficiently to permit restart.
PE-3	DC line fault	A fault on the DC transmission line or cable — from lightning, insulator flashover, conductor contact, or physical damage — initiates an automatic protection response that deenergizes the faulted segment. For bipolar systems, a pole fault may allow continued operation at half capacity on the healthy pole.	Monopolar operation following a pole fault reduces transfer capability by approximately half. The AC systems must manage the partial transfer reduction. Full DC line faults that require deenergization of both poles produce TCI equal to DC line repair time plus control system restart — potentially days to weeks depending on fault location and access.
PE-4	Loss of converter station auxiliary power	Loss of AC auxiliary power at either converter station removes the station service supply for cooling systems, valve hall ventilation, control electronics, and communications equipment. Converters must be blocked to prevent thermal damage. Equivalent to station service (hotel load) loss at FC2 facilities.	Station service loss at a converter station is the FC3 analog of PE-SS in DBA-ES-TX-FC2. The converter station cannot operate without continuous auxiliary power for cooling and control. Recovery requires restoration of auxiliary power before any converter restart attempt.

PE-5	AC network reconfiguration produces an inadmissible operating point	A change in AC network topology — from a fault, a planned switching operation, or emergency reconfiguration — moves the AC system at one or both converter terminals into an operating condition outside the HVDC control system’s stable operating range. The control system may be unable to maintain DC power transfer under the new AC conditions.	This is a system interaction event unique to HVDC: the HVDC link’s operating envelope is defined by AC system conditions at both terminals simultaneously. A change at the remote terminal can force a block at the local terminal without any physical event at the local facility.
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3.3 Category SE — Structural and Physical Integrity Events

Structural and physical integrity events at FC3 concentrate on the converter transformer as the governing long-lead-time element, and on the thyristor or IGBT valve assemblies as the secondary physical vulnerability. The geomagnetic disturbance event (SE-4) is unique to FC3 within the DBA-ES series: HVDC systems are the electric sector facility class most susceptible to GIC effects because of their long DC line lengths, grounded electrode configurations, and the GIC vulnerability of converter transformer winding configurations.

Code	Event	Initiating Condition	Governing Concern
SE-1	Converter transformer failure — internal fault	Internal fault within the converter transformer winding or bushing, resulting in differential relay operation and transformer removal from service. Converter transformers are custom-designed for HVDC service with specific tap changer, impedance, and insulation requirements that distinguish them from standard power transformers.	RT_HVLLC governs. Converter transformers are custom-manufactured components with lead times of twelve to eighteen months under normal procurement. Unlike standard EHV transformers, converter transformers cannot be sourced from the DOE/NERC STEP program inventory. There is no established strategic reserve.
SE-2	Converter transformer failure — external physical damage	Physical damage to the converter transformer from fire, flooding, projectiles, or blasts. Converter transformers are located within the converter station yard and are exposed to the same external physical attack vectors as EHV substations. Oil-filled tank and external bushings are the primary vulnerable elements.	This is the bounding SE event for FC3, combining the RT_HVLLC consequence of converter transformer replacement with the TCI consequence of link unavailability during replacement. The compound metric — TCI duration equal to RT_HVLLC — is the worst-case consequence for a single converter station attack.
SE-3	Thyristor valve damage or failure	Physical or electrical damage to the thyristor valve assemblies in the converter hall. Thyristor valves are the switching elements that perform the AC-DC conversion. They are housed in a	Thyristor valve replacement is faster than converter transformer replacement — lead times are typically four to eight weeks for standard valve assemblies — but

		dedicated valve hall structure with specific environmental controls. Valve damage may affect a single valve group or the entire converter.	requires specialized contractor capability and factory support. Physical damage to the valve hall structure that compromises environmental controls extends the repair timeline.
SE-4	Natural hazard — geomagnetic disturbance (GMD)	Severe geomagnetic disturbance induces geomagnetically induced currents (GIC) in long DC transmission lines and in the converter transformers at each terminal. GIC causes transformer half-cycle saturation, reactive power absorption, and potential thermal damage to converter transformer windings. HVDC systems are particularly susceptible to GIC relative to AC transmission because of their long line lengths and grounded electrode configurations.	A Carrington-class GMD event is the governing natural hazard for FC3. HVDC converter transformers are more vulnerable to GIC thermal damage than standard AC transformers because of their specialized winding configurations. A GMD event that damages converter transformers at both terminals simultaneously produces TCI equal to RT_HVLLC at each terminal, potentially extending the cross-interconnection isolation duration beyond what single-terminal damage would produce.
SE-5	Natural hazard — compound event	Sequential or simultaneous occurrence of two natural hazard events where recovery from the first is impaired by the second. The governing compound case for FC3 is a geomagnetic disturbance affecting the converter transformer condition, followed by a seismic event at one terminal that damages the weakened transformers beyond repair. The GMD-seismic sequence exploits the reduced dielectric and thermal margins of a GMD-stressed transformer.	The GMD-seismic compound case establishes the upper bound of the non-adversarial extended consequence regime for FC3. Under MA conditions, an adversary aware of a facility's post-GMD degraded state could exploit the reduced margins with a precision physical attack requiring less destructive energy than would be required against a fully conditioned transformer.

3.4 Category CE — Control and Communications Events

Control and communications events at FC3 have a structural characteristic that distinguishes them from CE events in other facility classes: HVDC control systems require continuous inter-terminal communication as a functional requirement of normal operation. This makes the communications link between converter stations a critical single-point dependency with no analog in AC transmission. CE events that sever or degrade inter-terminal communications do not merely reduce operational visibility — they may directly force a converter block or restrict the link to a degraded operating mode.

Code	Event	Initiating Condition	Governing Concern
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CE-1	Loss of HVDC control system communications	Loss of the telecommunications link between the rectifier and inverter converter stations. HVDC control systems require continuous high-speed communications between terminals to coordinate firing angles, manage power order changes, and execute protection functions. Communication loss forces the control system into a degraded operating mode or initiates automatic block.	Unlike AC transmission, HVDC requires active inter-terminal communication for normal operation. A communications failure is not merely an operational inconvenience — it may force a link block or limit the link to a reduced, fixed-power operating mode. The communications link is therefore a critical single-point dependency with no parallel in AC transmission.
CE-2	HVDC control system malfunction or cyber intrusion	Malfunction of or unauthorized access to the HVDC control system, including power order manipulation, firing angle modification, or protection relay setting changes. HVDC control systems are highly specialized digital systems with proprietary protocols. Unauthorized modification can cause immediate converter block or force the link into an operating mode that stresses AC system stability.	HVDC control systems are a high-value cyber target because manipulation can produce large, instantaneous power flow changes in the AC network without physical equipment damage. A forced power reversal or rapid power order change induced by cyber intrusion can destabilize AC systems at both terminals simultaneously.
CE-3	Loss of CntlCo situational awareness for DC power flow	Loss of SCADA visibility into HVDC link status, power flow, and converter station condition at the reliability coordinator or balancing authority level. The CntlCo manages AC system constraints in part by accounting for the HVDC link's contribution to transfer capability. Loss of DC power flow data degrades contingency analysis accuracy for the entire control area.	CE-3 does not block the HVDC link but removes the CntlCo's ability to correctly account for DC transfer capability in AC system management decisions. Incorrect contingency analysis can lead to operating the AC system in a configuration that would be unstable following an actual HVDC block event.

3.5 Category XC — Cross-Sector Cascade Events

Cross-sector cascade events at FC3 operate at a geographic scale that exceeds all other facility classes in the DBA-ES series. While FC2 (EHV substation) cascade events affect a metropolitan load pocket, FC3 cascade events affect an entire interconnection or large sub-region. The TCI metric is the defining characteristic of FC3's XC analysis: the duration of cross-interconnection transfer capability unavailability governs the magnitude of the cascade consequence.

XC-4 is identified at the DB layer — not only in the MA extension — because the timing dependency that an adversary could exploit is a structural characteristic of the HVDC facility class. Any event that produces a large TCI has different consequence severity depending on when in the dispatch cycle it occurs. This timing dependency must be part of the non-adversarial design basis because even non-adversarial events — a converter transformer failure during peak

import season — can produce consequences that the timing-agnostic design basis would underestimate.

Code	Event	Initiating Condition	Governing Concern
XC-1	Regional isolation through cross-interconnection transfer loss	Loss of the HVDC link removes the transfer capability that allows one interconnection to support another under stress. A region that was importing power through the HVDC link must source that power from within its own interconnection or curtail load. A region that was exporting loses the corresponding generation revenue and balancing flexibility.	This is the defining XC event for FC3 and the primary rationale for its placement in the DBA-ES series. The 2003 Northeast Blackout demonstrated how the loss of transfer capability between regions transforms a local event into a cross-regional cascade. The TCI metric is defined specifically to bound this consequence.
XC-2	Frequency instability in the islanded receiving region	An interconnection or sub-region that was importing a large block of HVDC power experiences an instantaneous generation deficit when the link blocks are removed. If the deficit exceeds the available frequency response in the receiving system, frequency declines below UFLS trigger thresholds, initiating automatic load shedding cascades.	The size of the frequency event is proportional to the HVDC transfer block size relative to the receiving system's total inertia and frequency response. For a 3,000 MW HVDC link blocking instantaneously into a receiving system with limited synchronous inertia, the frequency transient can be severe enough to initiate cascading UFLS before any operator action is possible.
XC-3	Cross-sector consequence through regional power deficit	Regional power deficit following HVDC block or extended TCI produces load shedding that removes power from cross-sector critical facilities in the receiving region. Defense-critical loads, water treatment, hospitals, and continuity-of-government sites in the region that lose HVDC import capability are the ultimate cross-sector consequence targets.	XC-3 is the Tier 3 pathway for FC3. Its consequence analysis must identify the critical load profile in the receiving region during peak import conditions and the backup power posture of the most time-constrained critical facilities. The Tier 3 acceptance criterion is defined by the most time-constrained critical facility in the affected region.
XC-4	Coordinated attack exploiting inter-regional energy market dependency.	An adversary who understands the energy market dependency between interconnections can time an HVDC attack to coincide with peak import dependency — the period when the receiving interconnection has committed to purchasing power that it cannot immediately replace from internal sources. The timing maximizes the consequence of the block event within the receiving system.	XC-4 is an MA-layer XC event introduced here at the DB layer because the timing dependency is a structural characteristic of the HVDC facility class that has no analog in AC transmission. It is not an equipment failure; it is a market and operational dependency that an adversary with intelligence about energy dispatch schedules can exploit.

3.6 Dual Design Basis Architecture

Consistent with DBA-ES §3.4, an FC3 facility may simultaneously carry a sector-specific physical Design Basis (this document) and an AI Governance Design Basis established under the AI Governance series. The dual-DB interaction at FC3 is most acute in two operational contexts.

HVDC control optimization. Modern HVDC control systems increasingly incorporate ML-based power flow optimization, voltage stability monitoring, and commutation failure prediction. These tools are ML systems operating below the Brightline — deterministic, formally testable — and may operate autonomously. However, their outputs govern the HVDC link’s power order and firing angle settings. A compromised ML optimization layer — whether through adversarial data poisoning or operation outside its training envelope during an unusual grid condition — can produce control outputs that destabilize the AC systems at both terminals simultaneously.

Anomaly detection for cyber intrusion. CE-2 identifies cyber intrusion into the HVDC control system as a postulated event. ML-based anomaly detection is the primary tool for identifying uncharacteristic control system behavior that may indicate intrusion. If the anomaly detection ML system is itself compromised or blinded — through adversarial manipulation of the training data or the sensor feeds it monitors — the primary cyber defense for the HVDC control system is removed. The design basis must address the adversarial ML attack on the anomaly detection layer as a CE-class event.

Section 4 — Design Basis Accidents: Non-Adversarial Envelope

4.1 Bounding Event Methodology

Bounding events for FC3 are identified through the methodology established in DBA-ES §4.1: physical plausibility, historical frequency, and consequence severity. For FC3, the bounding event is determined by two independent criteria: the event that produces the longest TCI, and the event that produces the most severe instantaneous AC system transient at the block. These two criteria may identify different events as bounding, and both must be addressed in the DBA.

The longest-TCI bounding event is SE-2 (converter transformer external physical damage requiring replacement), which produces TCI equal to RT_HVLLC of twelve to eighteen months. This is the design basis accident for FC3 from a consequence-duration perspective.

The most severe instantaneous transient bounding event is PE-1 (AC system fault causing forced converter block) at maximum transfer rate, which produces the largest instantaneous power flow change in the receiving AC system. This is the design basis accident for FC3 from a frequency stability perspective. Both bounding events must be analyzed separately.

4.2 Non-Adversarial Consequence Envelope

Tier 1 — Facility-level consequences. The bounding Tier 1 consequence is loss of HVDC transfer capability for a duration equal to RT_HVLLC (twelve to eighteen months) following converter transformer replacement-class damage. Monopolar operation at half capacity may be available if only one pole’s converter transformer is damaged.

Tier 2 — Grid-level consequences. The bounding Tier 2 consequence is the instantaneous frequency and voltage transient in the receiving AC system at the maximum transfer block, followed by a sustained regional power deficit for the duration of TCI. The instantaneous transient is bounded by the receiving system’s frequency response capability and UFLS relay settings. The sustained deficit is bounded by the receiving system’s resource adequacy without HVDC import contribution.

Tier 3 — Cross-sector consequences. The bounding Tier 3 consequence is load shedding in the receiving region during peak import dependency, removing power from defense-critical and other critical facilities for a duration determined by TCI and the critical facilities’ backup power posture. The Tier 3 acceptance criterion is defined by the most time-constrained critical facility in the receiving region.

4.3 Extended Consequence Regime

The extended consequence regime for FC3 is defined by the intersection of TCI and the receiving region’s resource adequacy posture during the TCI period. A TCI of twelve to eighteen months at maximum covers at least one full peak demand season in the receiving region. Resource adequacy planning must demonstrate that the receiving interconnection can sustain its reliability obligations without HVDC import contribution for this duration.

The extended consequence regime analysis must also address supply chain interdiction risk for converter transformer procurement. Unlike large AC power transformers — which have a limited domestic manufacturing base but at least several domestic manufacturers — HVDC converter transformers are manufactured by a very small number of global suppliers, none of which are domestic US manufacturers for units at the relevant voltage and power class. An adversary targeting the supply chain for converter transformer replacement can extend TCI beyond the nominal RT_HVLLC.

Section 5 — Military Action Extension

5.1 MA Extension Applicability

All FC3 facilities satisfy at least three of the six MA extension trigger criteria established in DBA-ES §1.5 by definition. Trigger 1 fires because the governing failure mode produces Tier 2 and Tier 3 consequences. Trigger 2 fires because HVDC interconnects serving cross-interconnection roles are designated as critical infrastructure under FERC and NERC oversight. Trigger 3 fires because converter transformers have $RT_HVLLC \geq 12$ months. Trigger 5

(cluster/corridor significance) also applies. All HVDC transfer capability in a given inter-regional corridor may be concentrated in a single link, making that link a corridor-level choke point. The MA extension is mandatory for all FC3 facilities.

5.2 Application of the Four Characteristics

The four MA characteristics are applied to each postulated event category in the governing scenarios table below.

Category	Governing MA Scenario	Application of Four Characteristics	TCI / RT_HVLLC Implication	Acceptance Criterion
PE	Coordinated a cyber-physical attack, forcing a sustained converter block at both terminals simultaneously	Intentionality: adversary targets the control system at both terminals to prevent automatic restart following the block. Persistence: control system compromise sustains the block beyond the normal recovery timeline. Precision: attack targets the inter-terminal communications link — the single dependency whose loss most effectively prevents restart. Denial: Physical access denial to converter stations prevents manual override of the control system block.	TCI begins at block initiation. Duration governed by control system recovery time under adversarial persistence — potentially days to weeks if physical access is denied. RT_HVLLC is not engaged unless physical equipment damage accompanies the control system attack.	Inter-terminal communications must have an independent, physically separated backup path that the same intrusion vector cannot compromise as the primary path.
SE	Precision kinetic attack on converter transformer at one or both terminals — designed to maximize RT_HVLLC	Intentionality: adversary selects converter transformer as the highest-RT_HVLLC element. Persistence: 12–18 months replacement lead time is the adversary’s effective persistence without further action; no STEP program equivalent exists for converter transformers. Precision: oil-filled bushings and external radiators are the specific vulnerable elements. Denial: Access denial to the converter station yard prevents damage assessment and replacement delivery.	TCI = RT_HVLLC = 12–18 months for converter transformer replacement. Both metrics are simultaneously engaged. This is the governing consequence case for FC3: cross-interconnection transfer capability is unavailable for the full RT_HVLLC duration.	Physical hardening of the converter transformer must reduce the probability of replacement-class damage to an acceptable level. Absence of a STEP equivalent for converter transformers is a strategic gap requiring national-level policy response.
CE	Cyber intrusion inducing forced power reversal or rapid uncontrolled	Intentionality: adversary studies HVDC control system architecture to identify the power order manipulation that produces	TCI during an active cyber event is determined by the operator’s ability to	HVDC control system must include anomaly

	power order change.	maximum AC system instability at the receiving terminal. Persistence: control system compromise persists until forensically identified and remediated. Precision: targets the specific control parameter whose manipulation causes AC frequency instability without an immediately obvious signature. Denial: communications backhaul damage removes SCADA visibility simultaneously.	detect anomalous behavior and manually block the link. Post-event TCI determined by control system forensic investigation and remediation timeline — potentially weeks.	detection for uncharacteristic power order changes or firing angle commands, with automatic protective block authority independent of the compromised control pathway.
XC	Attack timed to peak import dependency — coordinated with energy market intelligence.	Intentionality: adversary times attack to the hour of maximum HVDC import dependency in the receiving system, maximizing the generation deficit that the AC system must absorb. Persistence: RT_HVLLC sustains the deficit through the peak demand season. Precision: attack timing is the primary precision element — the same physical attack produces substantially larger consequences at peak import than at off-peak. Denial: Access denial prevents emergency generator dispatch coordination.	TCI consequence severity is a function of both duration and timing. An 18-month TCI during a peak summer import season produces qualitatively more severe regional consequences than the same TCI duration beginning in a low-demand period.	Regional resource adequacy planning must treat HVDC import capability as a contingency resource, not a firm resource, and demonstrate adequacy without HVDC contribution for the duration equal to RT_HVLLC.

5.3 The Regional Isolation Strategy

The governing MA scenario for FC3 is qualitatively different from any scenario in FC2. At FC2, the adversary’s objective is to deprive a metropolitan load pocket of power. At FC3, the adversary’s objective is to isolate a region from its external support infrastructure — to remove the interconnection’s ability to call upon neighboring systems for support under stress, and to remove the neighboring system’s ability to provide it.

Regional isolation through HVDC targeting is a strategic military objective, not a tactical one. It does not require attacking every generator in the target region. It requires removing the transfer capability that the region depends on for reliability under stress, at a time when the region is most dependent on that capability. An adversary who understands the energy market dependency between interconnections — who knows which hours the target region imports heavily from its neighbors, and which generating resources in the neighboring region are serving those exports — has the intelligence required to time an FC3 attack for maximum consequence.

This strategic character is the reason XC-4 (coordinated attack exploiting inter-regional energy market dependency) is identified at the DB layer rather than only in the MA extension. The timing dependency is a structural vulnerability of the HVDC facility class that exists independent of adversarial intent. Non-adversarial events that occur at peak import dependency already demonstrate this: a converter transformer failure in July during peak summer import produces worse consequences than the same failure in April. The MA extension adds intentionality to the timing — the adversary chooses July — but the underlying vulnerability is in the non-adversarial design basis.

5.4 Supply Chain as a Persistence Vector

The absence of a strategic reserve for HVDC converter transformers and the concentration of global manufacturing capacity in a small number of non-domestic suppliers creates a supply chain persistence vector that has no equivalent in other facility classes. An adversary who attacks an HVDC converter transformer and simultaneously interdicts the supply chain for replacement units extends TCI beyond the nominal RT_HVLLC without any further direct action against the facility.

Supply chain interdiction as an MA persistence vector is not hypothetical. The concentration of converter transformer manufacturing in a small number of global facilities — most of which are located outside the United States — creates discrete, identifiable choke points in the replacement supply chain. An adversary with the capability to execute a precision strike on an HVDC converter station almost certainly has the capability to surveil and, if desired, disrupt the logistics chain for replacement equipment. The FC3 design basis must treat supply chain interdiction as a component of the persistence characteristic, not an independent contingency.

Section 6 — Consequence Analysis

6.1 Consequence Chain: SE-2 / TCI-Maximum Governing Case

The governing consequence chain for FC3 proceeds as follows.

Initiating event. A precision physical attack on the converter transformer at one or both converter stations damages the external oil system, initiating transformer shutdown. Protection systems de-energize the affected converter. If only one terminal is attacked, the remaining terminal may sustain monopolar operation at reduced capacity depending on system configuration.

Immediate facility consequence (Tier 1). The attacked converter transformer is removed from service. HVDC transfer capability is reduced to zero (bipolar attack) or approximately fifty percent (monopolar attack). TCI begins. Converter transformer replacement is initiated; RT_HVLLC of twelve to eighteen months governs the restoration timeline.

Grid consequence (Tier 2). The receiving AC system experiences an instantaneous power deficit equal to the lost HVDC transfer block. Automatic generation control and governor response

begin frequency recovery. If the deficit exceeds the frequency response capability, UFLS relays operate to shed load. Over the subsequent hours, the system operator dispatches replacement generation from within the receiving interconnection. If sufficient replacement capacity is not available, controlled load shedding continues. For the duration of TCI, the receiving interconnection operates without its HVDC import resource.

Cross-sector consequence (Tier 3). Load shedding associated with the immediate frequency event and with sustained resource inadequacy during TCI removes power from critical facilities in the receiving region. The consequence escalates from temporary outage (resolved by backup generation) to sustained deficit (requiring fuel resupply and emergency imports from more distant sources) to potential mission degradation if TCI extends beyond backup generation endurance and no alternative supply is available.

Extended consequence regime. The receiving interconnection manages reliability obligations without HVDC import contribution for the TCI duration. Resource adequacy margins are reduced. Contingency management becomes more constrained. The risk of cascading following subsequent contingency events is elevated for the full TCI period.

6.2 MA Extension Consequence Chain

The MA extension consequence chain for FC3 differs from the non-adversarial chain in the following ways, corresponding to the four characteristics.

Intentionality extends the consequence from a single-terminal event to a coordinated attack on both terminals simultaneously, eliminating monopolar fallback. The adversary also times the attack to peak import dependency, maximizing the immediate frequency transient and the sustained deficit.

Persistence is provided by supply chain interdiction, extending TCI beyond RT_HVLLC. The adversary does not need to sustain physical presence at the converter station to extend the consequence — supply chain disruption provides persistence at a geographic remove from the attacked facility.

Precision targets the converter transformer specifically — the element whose replacement produces the longest TCI — rather than the DC line (which is repairable in days to weeks) or the thyristor valves (repairable in weeks). Precision also targets the attack timing to peak import dependency.

Denial prevents damage assessment, delays replacement equipment delivery to the converter station yard, and may extend to area denial that prevents contractor mobilization for the replacement installation. Access denial under MA conditions can extend the effective RT_HVLLC beyond the nominal procurement lead time.

Section 7 — Acceptance Criteria

7.1 Tier 1 Acceptance Criteria

The facility must be capable of sustaining monopolar operation following the loss of one pole's converter transformer, maintaining at least fifty percent of the design transfer capacity during the monopolar period. Station service continuity must be maintained for a minimum of seventy-two hours following loss of all AC supply through UPS or dedicated auxiliary generation.

Converter transformer physical hardening must reduce the probability of replacement-class external damage from the design basis physical attack to an acceptable level. Acceptance criteria for physical hardening are dependent on site-specific threat assessment and are not fully definable at the series level.

7.2 Tier 2 Acceptance Criteria

The receiving AC system must demonstrate, through NERC TPL planning analysis, that it can maintain reliability obligations following instantaneous loss of the HVDC link at maximum transfer rate under all pre-contingency operating conditions within the TPL planning envelope. Frequency deviation following a maximum-transfer block event must not exceed the UFLS relay trigger threshold that would initiate load shedding beyond the NERC-allowable limits for the planning event class.

The receiving interconnection must demonstrate resource adequacy without HVDC import contribution for a duration equal to the maximum credible TCI — defined for planning purposes as RT_HVLLC (twelve to eighteen months) — for all seasonal demand periods. Resource adequacy assessment must treat HVDC import as a contingency resource, not a firm resource, consistent with the TCI consequence duration.

7.3 Tier 3 Acceptance Criteria

The maximum duration of power interruption to any defense-critical or other highest-priority critical facility in the receiving region must not exceed that facility's backup power posture duration before mission-unrecoverable consequences occur. This criterion requires a regional critical load inventory for the receiving interconnection.

Where the most time-constrained critical facility in the receiving region has a backup power posture shorter than the maximum credible TCI, the Tier 3 criterion cannot be satisfied by grid restoration alone. In this case, the criterion must be satisfied through alternative supply arrangements — dedicated mobile generation contracts, pre-positioned fuel reserves, or priority restoration agreements — that are established before an FC3 event occurs and that remain available under the access denial conditions the MA extension postulates.

Section 8 — Deferred Items

Item	Description	Resolution Criterion
DI-1	Multi-terminal HVDC scope. This document addresses point-to-point HVDC as the primary case. Multi-terminal HVDC systems — in which three or more converter stations share a common DC bus — present additional complexity: a fault or attack at one terminal affects all other terminals simultaneously, and the control coordination across multiple terminals requires a more complex protective architecture. Multi-terminal HVDC is deferred as a scope extension pending development of the point-to-point analysis.	Multi-terminal HVDC scope extension added to FC3 instrument or addressed in a dedicated annex following completion of the point-to-point analysis.
DI-2	Converter transformer strategic reserve gap. Unlike large power transformers for AC substations, converter transformers are not covered by the DOE/NERC STEP program. There is no national strategic reserve for HVDC converter transformers. This gap has no institutional owner and no resolution pathway currently identified. A policy analysis identifying the appropriate federal agency to own this gap and a cost-benefit framework for establishing a limited converter transformer reserve is needed.	Policy analysis completed; federal agency ownership identified; cost-benefit framework for strategic reserve established and presented to DOE and FERC.
DI-3	TCI acceptance criterion calibration. Section 7.3 establishes TCI as the governing consequence-duration metric and identifies regional resource adequacy as the acceptance criterion framework. The specific TCI duration threshold above which regional resource adequacy cannot be maintained without unrecoverable cross-sector consequences requires quantitative analysis against each FC3 facility's specific receiving-system load profile and critical facility inventory. This quantitative calibration is facility-specific and cannot be completed at the series level.	TCI acceptance criterion calibrated for each FC3 facility as part of site-specific design basis implementation, following guidance to be developed in a companion implementation guide.
DI-4	GMD postulated event quantification. SE-4 identifies geomagnetic disturbance as the governing natural hazard for FC3 and the Carrington-class event as the bounding case. Quantitative GIC analysis for specific HVDC converter transformer configurations requires site-specific geomagnetic latitude data, soil conductivity models, and DC resistance parameters for the converter transformer windings. This analysis is facility-specific and is not completed here.	Quantitative GIC analysis completed for each FC3 facility as part of site-specific design basis implementation, consistent with NERC GMD standards TPL-007 and related guidance.
DI-5	Dual-DB AI Governance interaction points. Section 3.6 acknowledges the dual-DB architecture per DBA-ES §3.4. The specific AI governance interaction points for FC3 — ML systems used in HVDC control optimization, power flow forecasting, and anomaly detection — are identified but not yet fully analyzed. The Brightline boundary between ML (autonomous) and Gen AI (command-broker-required) applications must be drawn explicitly for FC3 control system AI contexts.	AI governance interaction points for FC3 control system AI applications identified and mapped to Brightline / Command Broker architecture at next revision.

Section 9 — References

Regulatory and Standards References

Federal Energy Regulatory Commission. Order No. 693: Mandatory Reliability Standards for the Bulk-Power System. Washington, D.C.: FERC, 2007.

North American Electric Reliability Corporation. FAC-001-3: Facility Connection Requirements. Atlanta: NERC. Current version.

North American Electric Reliability Corporation. FAC-002-3: Facility Connection Requirements. Atlanta: NERC. Current version.

North American Electric Reliability Corporation. TPL-001-5: Transmission System Planning Performance Requirements. Atlanta: NERC. Current version.

North American Electric Reliability Corporation. TPL-007-4: Transmission System Planned Performance for Geomagnetic Disturbance Events. Atlanta: NERC. Current version.

North American Electric Reliability Corporation. CIP-014-3: Physical Security. Atlanta: NERC. Current version.

Government and Policy References

Defense Science Board Task Force on DoD Energy Strategy. “More Fight — Less Fuel.” Washington, D.C.: Office of the Under Secretary of Defense for Acquisition, Technology and Logistics, February 2008.

National Infrastructure Protection Plan 2013: Partnering for Critical Infrastructure Security and Resilience. Washington, D.C.: DHS, 2013.

U.S. Department of Energy and North American Electric Reliability Corporation. “Final Report on the August 14, 2003 Blackout in the United States and Canada: Causes and Recommendations.” Washington, D.C.: DOE, April 2004.

Technical References

CIGRE Working Group B4.62. “Connectionless DC Grids.” CIGRE Technical Brochure 533. Paris: CIGRE, 2013.

CIGRE Working Group B4.07. “Guide to the Specification and Design Evaluation of AC Filters for HVDC Systems.” CIGRE Technical Brochure 139. Paris: CIGRE, 1999.

Manitoba Hydro. “Bipole III Reliability Project.” Project documentation, Manitoba Hydro, Winnipeg, 2021.

Mohan, Ned, Tore M. Undeland, and William P. Robbins. Power Electronics: Converters, Applications, and Design. 3rd ed. Hoboken: John Wiley & Sons, 2003. (Reference for converter commutation failure analysis.)

Series Parent and Cross-Reference Documents

Roxey, Tim. DBA-ES: Electric Sector Design Basis Framework. Version 0.6 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Tim. DBA-ES-TX-FC2: Large EHV Transmission Substation. Version 0.1 — DRAFT. Eclectic Technologies, June 2026.

Roxey, Tim. DBA-ES-HY-FC1: Large Hydroelectric Storage Facilities. Version 0.3 — DRAFT. Eclectic Technologies, May 2026.

Roxey, Tim. GenMix Optimization Whitepaper. Version 0.3 — DRAFT. Eclectic Technologies, 2026. (Origin of RT_HVLLC construct.)